Stefano Roy Bisignano

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AI Systems Engineer specializing in **LLM architectures**, **edge inference**, and **Physical AI**. Applying rigorous **LLMOps**, I build end-to-end systems optimized for inference efficiency, low latency, and robust automation under real-world industrial and computational constraints.

Personal: I rely on obsessive perseverance rather than quick talent. When a problem clicks, I dive into it down to its core mechanics. Outside of work, I bring this same trial-by-fire dedication to surfing and freestyle skiing.

Experience

ML Engineer → ML Engineer Lead (promoted) **May 2024 – Oct 2025**
Simplex Rapid *Milan, Italy*

- Acted as the core technical reference for AI infrastructure; structured end-to-end ML pipelines and coordinated developer resources to align research with manufacturing constraints.
- Built a multilingual translation pipeline with fine-tuned models across **20 languages**; achieved **~53% touchless rate** and reduced human effort from **~8,000h** to **~2,350h** (**~71%** reduction).
- Migrated an agentic business assistant to the Responses API with tool calling, monitoring, and tracing for reliability and auditability.
- Combined LLM and vision to extract technical parameters from spring drawings/PDFs with per-field validation and human-in-the-loop review, enabling automatic population of manufacturing data.

Projects

Edge-VLA-Micro. *Built a voice-controlled FPV drone agent: camera + speech input feed a VLM action proposal, symbolic HSV/Pydantic guards block hallucinated moves, and validated commands execute through MAVSDK/PX4; optimized TTFT by 65.6%.* **2026**

Humanoid-Motion-Diffusion. *Audio-conditioned DDPM Transformer for SMPL/AIST++ humanoid motion; biomechanical evaluation reduced TSI 12.60 to 0.08.* **2026**

SSL for RCC Classification. *Self-supervised learning on histopathology WSIs for renal carcinoma subtype discovery* ***awarded full marks with honors (30/30 cum laude)****; research collaboration currently being extended toward an international publication.* **2026**

Embedded Vision Trade-offs (Arduino Nicla Vision). *Hardware-in-the-loop benchmark suite exposing deployment trade-offs beyond validation accuracy; protocol-driven evaluation and reproducible runs.* **2026**

Affordance 3D Highlighting. *Zero-shot localization of functional 3D regions from language prompts, inspired by CLIP and 3D Highlighter (CVPR 2023).* **2025**

Swiss AI Weeks (ETH AI Center 2025). *Multi-agent dialog-to-action system with disagreement arbitration and safe human handoff for financial advisory tasks.* **2025**

OnlyFly (HackUPC 2025, 4th/150+). *AI home-exchange MVP integrating preference matching, flight search (Skyscanner), and payments (Revolut).* **2025**

PostGenius (GenAI.Works 2024, 7th/4,500). *RAG-driven multimedia generator (text, image, video) built with OpenAI, Vectara, and Runway APIs.* **2024**

Education

MEng in Artificial Intelligence and Data Analytics, Politecnico di Torino, Italy **Apr 2024 – Expected Dec 2026**

BSc in Computer Engineering, Politecnico di Torino, Italy **Oct 2020 – Apr 2024**

Skills & Interests

AI Systems & Opt:	Inference Optimization, Model Quantization, LLM Architectures, Multi-Agent Systems
Programming:	Python, C/C++, Rust, CUDA, Bash, PyTorch, JAX, Java
Robotics & Sim:	ROS2, MuJoCo, Gymnasium, Gazebo, Physics-Informed Simulations, NVIDIA Jetson, 3D-GS
Infra & MLOps:	Docker, Kubernetes, CI/CD Pipelines, FastAPI, Triton Inference Server, Production Monitoring
Languages:	Italian (Native), English (Fluent), German (Basic), Spanish (Basic)
Interests:	Cognitive Psychology, Political Philosophy, Mechanical Inventions, Freestyle Skiing